

2.2 Bilinear Transformation

Def: The transformation $w = f(z) = \frac{az+b}{cz+d}$, where a, b, c, d are complex constants and $ad-bc \neq 0$, is called a Bilinear transformation (or, Mobius transformation).

Note: ① The inverse of the bilinear transformation is also a bilinear transformation.

② Let $w = \frac{az+b}{cz+d}$ be a bilinear transformation. Then the roots of $z = \frac{az+b}{cz+d}$ are called fixed points or invariant points of the bilinear transformation $w = \frac{az+b}{cz+d}$.

③ The cross ratio of four points z_1, z_2, z_3 and z_4 is given by

$$(z_1, z_2, z_3, z_4) = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_2 - z_3)(z_4 - z_1)}$$

e.g: let $w = \frac{z-2}{2z+1}$ be a bilinear transformation.

$$\begin{aligned} \text{Now, } z = \frac{z-2}{2z+1} &\Rightarrow 2z^2 + z = z - 2 \\ &\Rightarrow 2(z^2 + 1) = 0 \\ &\Rightarrow z = -i, i \end{aligned}$$

Therefore, $z = -i, i$ are the invariant points of the bilinear transformation $w = \frac{z-2}{2z+1}$.

⑤ If z_1, z_2, z_3, z_4 are four points and w_1, w_2, w_3, w_4 are the corresponding images under the bilinear transformation $w = \frac{az+b}{cz+d}$, then

$$(z_1, z_2, z_3, z_4) = (w_1, w_2, w_3, w_4)$$

i.e., The bilinear transformation preserves the cross ratio.

⑥ The identity map $w = z \left(= \frac{1 \cdot z + 0}{0 \cdot z + 1} \right)$ is a bilinear transformation.

⑦ Bilinear transformation is also called as the Examples: linear fractional transformation.

1. Show that the transformation $w = \frac{2z+3}{z-4}$ maps the circle $x^2 + y^2 - 4x = 0$ onto the line $4u+3=0$ (here $w = u+iv$)

Sol: Given $w = \frac{2z+3}{z-4}$ or, $z = \frac{4w+3}{w-2}$

$$\text{Now, } x^2 + y^2 - 4x = 0$$

$$\Rightarrow |z|^2 - 4 \operatorname{Re} z = 0$$

$$\Rightarrow z \cdot \bar{z} - 4(z + \bar{z}) = 0$$

$$\Rightarrow \left(\frac{4w+3}{w-2} \right) \left(\frac{4\bar{w}+3}{\bar{w}-2} \right) - 4 \left(\frac{4w+3}{w-2} + \frac{4\bar{w}+3}{\bar{w}-2} \right) = 0$$

$$\Rightarrow (4w+3)(4\bar{w}+3) - 2[4w+3)(\bar{w}-2) + (4\bar{w}+3)(w-2)] = 0$$

$$\Rightarrow 2(w+\bar{w})+3=0 \Rightarrow 4u+3=0 \text{ which is the required line.}$$

Let $z = x+iy$
 then $|z|^2 = x^2 + y^2$,
 $\operatorname{Re} z = x, \operatorname{Im} z = y$

② show that the transformation $w = \frac{z-i}{z+i}$ maps the real axisⁱⁿ the z -plane into the unit circle $|w|=1$ in the w -plane.

Sol: Given $w = \frac{z-i}{z+i}$

or, $z = \frac{-i(w+1)}{w-1}$

$$x+iy = \frac{-i(u+iv+1)}{u+iv-1}$$

$$= \frac{v - i(u+1)}{(u-1)+iv} \cdot \frac{[(u-1)-iv]}{[(u-1)-iv]}$$

$$= \frac{v(u-1) - iv^2 - i(u^2-1) - v(u+1)}{(u-1)^2 + v^2}$$

$$= \frac{-2v}{(u-1)^2 + v^2} + i \frac{(1-u^2) - v^2}{(u-1)^2 + v^2}$$

$$\therefore x = \frac{-2v}{(u-1)^2 + v^2} \quad \text{and} \quad y = \frac{(1-u^2) - v^2}{(u-1)^2 + v^2}$$

Now, $y=0 \Rightarrow 1-u^2-v^2=0$

$$\Rightarrow u^2 + v^2 = 1$$

or, $|w|=1$

③ Show that the line $x=3y$ is mapped onto the circle under the bilinear transformation

$$w = \frac{iz+2}{4z+i}. \text{ Also, find the centre and radius}$$

of the image circle.

Answer: $(-3/8, -7/8)$ and radius $\frac{3}{4}\sqrt{\frac{5}{2}}$ units.

④ Find the bilinear transformation which maps the points $(-1, 0, 1)$ into the points $(0, i, 3i)$.

Sol: Let $z_1 = -1, z_2 = 0, z_3 = 1, z_4 = z$
and $w_1 = 0, w_2 = i, w_3 = 3i, w_4 = w$

By the cross ratio of the four points,
we have

$$\frac{(z_1 - z_2)(z_3 - z_4)}{(z_2 - z_3)(z_4 - z_1)} = \frac{(w_1 - w_2)(w_3 - w_4)}{(w_2 - w_3)(w_4 - w_1)}$$

$$\Rightarrow \frac{(-1-0)(1-z)}{(0-1)(z+1)} = \frac{(0-i)(3i-w)}{(i-3i)(w-0)}$$

$$\Rightarrow \frac{(z-1)}{-(z+1)} = \frac{3i-w}{2w}$$

$$\Rightarrow (z-1)2w = -(z+1)(3i-w)$$

$$\Rightarrow (z-1)2w = -3zi + zw - 3i + w$$

$$\Rightarrow (z-1)2w - zw - w = -3(z+1)i$$

$$\Rightarrow (2z-2-z-1)w = -3(z+1)i$$

$$\Rightarrow (z-3)w = -3i(z+1)$$

$$\Rightarrow w = \frac{3i(z+1)}{3-z}$$

or, $w = \frac{3iz + 3i}{3-z}$ which is the

required bilinear transformation.

- ⑤ Find the bilinear transformation which maps the points $(1, -i, 2)$ onto the points $(0, 2, -i)$.

Answer: The required bilinear transformation

is $w = \frac{2iz - 2i}{(i-1)z - 2i}$